

E-001 COVER / LEGEND / DEVICE SCHEDULE

Conv_Simple / CV-101 (garage conveyor) — device schedule · sheet index · wire-numbering key · line-style legend

DEVICE SCHEDULE

Tag	Type	Model	Role	Evidence
PLC1	plc	Allen-Bradley Micro820 2080-LC20-20QBB	Conveyor controller; Modbus RTU master to VFD1	verified
VFD1	vid	AutomationDirect GS10 DURApulse	Drives conveyor motor. Power input is 230 V single-phase (R/L1, S/L2; T/L3 un...	verified
M1	motor		Conveyor drive motor — 230 V 3~ from VFD1 U/V/W (voltage technician-confirmed...	field_verify
S0	e stop		Emergency stop, DUAL-CHANNEL (diverse) — NC ch to I-02, NO ch to I-03	verified
SS1	selector switch		Direction selector — FWD contact to I-00, REV contact to I-01 (FWD-OFF-REV)	verified
S2	pushbutton (NO)		Run / rearm pushbutton to I-04; illuminated — lamp fed by PLC O-03 (PBRunLED)	verified
B1	photo eye		Product-present beam to I-05 (blocked -> pe_latched soft stop)	verified
PS1	power supply	Mean Well DIN-rail supply (24 V / 1.0 A; 100-24...	24 VDC control supply (E-004). Feeds the DC +/- distribution block (blue = on...	verified
DB1	distribution block		24 VDC +/- distribution block (push-in, WAGO-style, 2-level), fed from PS1's ...	verified
CB1	circuit breaker		2-pole branch breaker on the 230 V single-phase supply (existence/type/rating...	field_verify
Q1	control relay	Schneider Electric TeSys CA3KN22BD (control rel...	Control relay (Schneider CA3KN22BD, hand-labeled 'MLC') USED AS THE VFD SUPPL...	verified
PL1	pilot light		Green RUNNING pilot — load of O-00 (LightGreen)	verified
PL2	pilot light		Red FAULT/E-STOP pilot — load of O-01 (LightRed)	verified
X1	terminal block		Field terminal strip — field devices land here before PLC (E-008)	field_verify

SHEET INDEX

Id	Title	Status
E-001	Cover / legend / device schedule	drafted
E-002	Power one-line	drafted
E-003	VFD power (CB1 -> VFD1 -> M1)	drafted
E-004	24 VDC control power distribution (PS1)	drafted
E-005	PLC digital inputs	drafted
E-006	PLC outputs (O-00..O-06)	drafted
E-007	RS-485 Modbus (PLC1 Ch2 <-> VFD1 RJ45)	drafted
E-008	Terminal strip (X1) + wire list	drafted
E-009	Open items / field verification	drafted

WIRE-NUMBERING KEY

Conductor numbers are W[sheet-digit][line-2d] (e.g. W503 = sheet E-005, line 3) — all PROPOSED until as-built (OI-03/OI-19). Rail/bus labels (W24, W0V) are exempt. Comms sheets (E-007) use mnemonic conductor labels (485+/485-/SGND/SH) per CommsToVFD ancestry. Same electrical node = same wire number; a number changes only through a device.

LINE-STYLE LEGEND

LEGEND

- VERIFIED (solid)
- FIELD VERIFY (dashed)
- Wxxxx

proposed wire number (red = unverified)

SAFETY

- LOTO + wait ≥5 min for DC-bus discharge before touching (WI p.2 §2).
- Monitored e-stop is NOT a safety stop — hardwire S0 to remove drive power per NFPA 79 / EN 60204-1 (style law §7; docs/references/industrial-wiring-diagram-standards.md).

NOTES

- LINE-STYLE LEGEND, red: red marks EITHER FIELD-VERIFY status (a dashed, unverified conductor/device or an unverified wire-number tag) OR a safety-critical prohibition (a NOT/never instruction inside a sheet's SAFETY block, e.g. 'a monitored input is NOT a safety stop') — same color, two distinct meanings distinguished by context (dashed line vs. SAFETY-block bullet). This is intentional package-wide convention, not per-sheet drift.

COVER / LEGEND / DEVICE SCHEDULE

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Drawn: MIRA / FactoryLM
cover generated from model/*.yaml

SHEET

E-001

REV A

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